

Assignment 3: End-to-End ML Versioning

Part A: Git Fundamentals & Advanced Commands

Student: Munam

Repository: github.com/Munam32/MLOps

Working branch: dev

A1 Repository setup

The project was initialised with `git init`. The first commit on `main` adds `README.md` and a `.gitignore` that excludes the virtual environment, Python caches, the DVC cache and `*.h5` model files (which DVC will track).

A1 · initial commit on main

```
$ git show --stat --format='commit %h (%D)%nAuthor: %an%n%n %s' $(git rev-list --max-parents=0 main)
commit 33fa36b (origin/main, origin/HEAD, main)
Author: Munam

Initial Commit

.gitignore | 4 ++++
README.md | 1 +
2 files changed, 5 insertions(+)

$ git show main:.gitignore
.venv/
__pycache__/
.dvc/cache/
*.h5
$ git branch
* dev
main
```

A2 · feature commits on dev

```
$ git rev-list --count main..dev
8

$ git log --oneline main..dev
85e40dd feat: add evaluate.py, metrics.json, confusion_matrix.png
bcbf0cc feat: create train.py
8e2f0da chore: modified params.yaml opening structure
6e3f3a6 feat: add params.yaml and preprocess.py
184b6fd chore: changing keras declaration
9901c46 feat: update .gitignore
7c201e3 feat: setup requirements.txt
7d60d03 feat: create prepare.py
```

A2 Feature branch

`dev` was created with `git checkout -b dev`. All Part B work happened there as small commits, one per script or config change: 8 commits ahead of `main` before Part A's scenarios began (shown above).

A3 Log variants

COMMAND	WHAT IT SHOWS THAT PLAIN <code>git log</code> DOES NOT
<code>--oneline --graph --all</code>	Every branch at once (not just the current one), one line per commit, with an ASCII graph of where branches split and merge.
<code>--stat -3</code>	For the last 3 commits, which files changed and how many lines were added or removed in each.
<code>-p -1</code>	The full patch of the latest commit: the actual lines added and removed, not just the message.
<code>main..dev</code>	Only commits reachable from <code>dev</code> but not <code>main</code> , meaning exactly what merging <code>dev</code> would bring in.

git log --oneline --graph --all

```
$ git log --oneline --graph --all
* 85e40dd feat: add evaluate.py, metrics.json, confusion_matrix.png
* bcbf0cc feat: create train.py
* 8e2f0da chore: modified params.yaml opening structure
* 6e3f3a6 feat: add params.yaml and preprocess.py
* 184b6fd chore: changing keras declaration
* 9901c46 feat: update .gitignore
* 7c201e3 feat: setup requirements.txt
* 7d60d03 feat: create prepare.py
* 33fa36b Initial Commit
```

git log --stat -3

```
$ git log --stat -3
commit 85e40dd3ee2c106fc2283d4ab65eefdfa9468640
Author: Munam32 <abdulmunam891@gmail.com>
Date: Wed Sep 30 12:51:37 2026 +0500

    feat: add evaluate.py, metrics.json, confusion_matrix.png

    confusion_matrix.png | Bin 0 -> 92050 bytes
    metrics.json         | 4 ++++
    src/evaluate.py      | 62 +++++
    +++++
    3 files changed, 66 insertions(+)

commit bcbf0cc772463e73a17dd900f96d1b9cf90932a4
Author: Munam <abdulmunam891@gmail.com>
```

```
git log -p -1
```

```
$ git log -p -1
```

```
commit 85e40dd3ee2c106fc2283d4ab65eefdfa9468640
Author: Munam32 <abdulmunam891@gmail.com>
Date: Wed Sep 30 12:51:37 2026 +0500
```

```
    feat: add evaluate.py, metrics.json, confusion_matrix.
    png
```

```
diff --git a/confusion_matrix.png b/confusion_matrix.png
new file mode 100644
```

```
index 00000000..54f06f5
```

```
Binary files /dev/null and b/confusion_matrix.png differ
```

```
diff --git a/metrics.json b/metrics.json
```

```
new file mode 100644
```

```
index 00000000..95543f5
```

```
--- /dev/null
```

```
+++ b/metrics.json
```

```
@@ -0,0 +1,4 @@
```

```
+{
```

```
+  "test_loss": 0.34686097502708435,
```

```
+  "test_accuracy": 0.8863999843597412
```

```
+}
```

```
\ No newline at end of file
```

```
diff --git a/src/evaluate.py b/src/evaluate.py
```

```
new file mode 100644
```

```
index 00000000..053ab45
```

```
--- /dev/null
```

```
+++ b/src/evaluate.py
```

```
@@ -0,0 +1,62 @@
```

```
+"""Evaluate the trained model on the test set, writing me-
+trics.json and a confusion matrix plot."""
```

```
+
```

```
+import json
```

```
+from pathlib import Path
```

```
+
```

```
+import keras
```

```
+import matplotlib
```

```
... 55 more lines omitted
```

```
Date: Mon Sep 28 14:42:14 2026 +0500
```

```
    feat: create train.py
```

```
.gitignore | 3 ++-
```

```
params.yaml | 7 ++++++
```

```
requirements.txt | 3 ++-
```

```
src/train.py | 52 ++++++
```

```
+++++
```

```
4 files changed, 63 insertions(+), 2 deletions(-)
```

```
commit 8e2f0da6c51d8580679d995a99b47bba8ac82f95
```

```
Author: Munam <abdulmunam891@gmail.com>
```

```
Date: Mon Sep 28 01:46:20 2026 +0500
```

```
    chore: modified params.yaml opening structure
```

```
src/preprocess.py | 3 ++-
```

```
... 1 more lines omitted
```

```
git log main..dev
```

```
$ git log main..dev
```

```
commit 85e40dd3ee2c106fc2283d4ab65eefdfa9468640
Author: Munam32 <abdulmunam891@gmail.com>
Date: Wed Sep 30 12:51:37 2026 +0500
```

```
    feat: add evaluate.py, metrics.json, confusion_matrix.
    png
```

```
commit bcbf0cc772463e73a17dd900f96d1b9cf90932a4
```

```
Author: Munam <abdulmunam891@gmail.com>
```

```
Date: Mon Sep 28 14:42:14 2026 +0500
```

```
    feat: create train.py
```

```
... 35 more lines omitted
```

A4 Diff variants

The README was expanded on `dev`. Before staging, `git diff` shows the change. After `git add`, plain `git diff` is empty and `git diff --staged` shows it instead, because it compares the staging area with the last commit.

(a) unstaged change

```
$ git status --short
```

```
M README.md
```

```
$ git diff
```

```
diff --git a/README.md b/README.md
```

```
index 3d0e341..c904c4f 100644
```

```
--- a/README.md
```

```
+++ b/README.md
```

```
@@ -1,1 +1,23 @@
```

```
-## MLOps Assignment 1
```

(b) after git add

```
$ git add README.md
```

```
$ git diff
```

```
$ git diff --staged --stat
```

```
README.md | 24 ++++++
```

```
1 file changed, 23 insertions(+), 1 deletion(-)
```

```

\ No newline at end of file
+## MLOps Assignment 1: Fashion-MNIST ANN Pipeline
+
+A fully connected neural network that classifies Fashion-
MNIST images into 10 clothing categories,
+versioned end to end with Git and DVC.
+
+## Pipeline
+
... 16 more lines omitted

```

```

$ git status --short
M README.md

```

Two-dot vs three-dot. `git diff main..dev` compares the two branch tips directly, so changes made on `main` since the branches split appear too, reversed. Here it shows `-## MLOps Assignment 3`: the hotfix from `A6` looks as if `dev` were undoing it. `git diff main...dev` compares the *merge base* (the commit where `dev` branched off) with `dev`, so it shows only what `dev` itself changed. That is what a pull request shows. The file statistics are identical here because the hotfix touched only a README line that `dev` also rewrote.

(c) two-dot vs three-dot, after the hotfix merge

```

$ git diff --stat main..dev
.gitignore      | 4 +++-
README.md       | 24 ++++++
confusion_matrix.png | Bin 0 -> 92050 bytes
metrics.json    | 4 ++++
params.yaml     | 10 ++++++
requirements.txt | 7 ++++++
scratch_inspect.py | 6 +++++
src/evaluate.py | 62 ++++++
src/prepare.py  | 27 ++++++
src/preprocess.py | 50 ++++++
src/train.py    | 52 ++++++
... 1 more lines omitted

$ git diff --stat main...dev
.gitignore      | 4 +++-
README.md       | 24 ++++++
confusion_matrix.png | Bin 0 -> 92050 bytes
metrics.json    | 4 ++++
params.yaml     | 10 ++++++
requirements.txt | 7 ++++++
scratch_inspect.py | 6 +++++
src/evaluate.py | 62 ++++++
src/prepare.py  | 27 ++++++
src/preprocess.py | 50 ++++++
src/train.py    | 52 ++++++
... 1 more lines omitted

$ git diff main..dev -- README.md | head -12
diff --git a/README.md b/README.md
index 2c64868..c904c4f 100644
--- a/README.md
+++ b/README.md
@@ -1,23 @@
-## MLOps Assignment 3
\ No newline at end of file
+## MLOps Assignment 1: Fashion-MNIST ANN Pipeline
+
+A fully connected neural network that classifies Fashion-MNIST images into 10 clothing categories,
+versioned end to end with Git and DVC.
... 1 more lines omitted

$ git diff main...dev -- README.md | head -12
diff --git a/README.md b/README.md
index 3d0e341..c904c4f 100644
--- a/README.md
+++ b/README.md
@@ -1,23 @@
-## MLOps Assignment 1
\ No newline at end of file
+## MLOps Assignment 1: Fashion-MNIST ANN Pipeline
+
+A fully connected neural network that classifies Fashion-MNIST images into 10 clothing categories,
+versioned end to end with Git and DVC.
... 1 more lines omitted

```

A5 Stash

While adding a pixel-range assertion to `preprocess.py` (uncommitted), I stashed the edit, switched to `main` to check its README, then came back and restored the work. `git stash list` shows the saved entry before `git stash pop` reapplies and drops it. Checking `main`'s README revealed the wrong assignment number, which is fixed in A6.

```
A5 · stash, switch, switch back, pop

$ git status --short
M src/preprocess.py

$ git stash push -m 'wip: pixel range check in preprocess'
Saved working directory and index state On dev: wip: pixel range check in preprocess

$ git status --short

$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

$ git log --oneline -1
33fa36b Initial Commit

$ cat README.md
## MLOps Assignment 1
$ git switch dev
Switched to branch 'dev'
Your branch is ahead of 'origin/dev' by 1 commit.
(use "git push" to publish your local commits)

$ git stash list
stash@{0}: On dev: wip: pixel range check in preprocess

$ git stash pop
On branch dev
Your branch is ahead of 'origin/dev' by 1 commit.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   src/preprocess.py

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (45d54c2fdc7713f4f6b30d8f1cd354be66153d3e)

$ git status --short
M src/preprocess.py
```

A6 Hotfix and rebase

A short-lived `hotfix` branch from `main` corrected the README title ("Assignment 1" → "Assignment 3"). It was merged into `main` with `--no-ff` so the merge is visible in the graph, then deleted.

```
A6 · hotfix branch merged into main

$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

$ git switch -c hotfix
Switched to a new branch 'hotfix'

$ sed -i 's/Assignment 1/Assignment 3/' README.md && git diff
diff --git a/README.md b/README.md
index 3d0e341..2c64868 100644
--- a/README.md
+++ b/README.md
@@ -1,1 @@
-## MLOps Assignment 1
\ No newline at end of file
+## MLOps Assignment 3
\ No newline at end of file

$ git commit -am 'fix: correct assignment number in README'
```

```
Before rebase

$ git log --oneline --graph --all
* dfdcbc3 chore: add scratch script to inspect raw arrays
* 2640bbc feat: assert normalized pixels stay within [0, 1]
* eb24041 docs: expand README with pipeline overview and setup
* 85e40dd feat: add evaluate.py, metrics.json, confusion_matrix.png
* bcbf0cc feat: create train.py
* 8e2f0da chore: modified params.yaml opening structure
* 6e3f3a6 feat: add params.yaml and preprocess.py
* 184b6fd chore: changing keras declaration
* 9901c46 feat: update .gitignore
* 7c201e3 feat: setup requirements.txt
* 7d60d03 feat: create prepare.py
| * bdd6d6f Merge branch 'hotfix'
|/|
| * 708ec94 fix: correct assignment number in README
```

```
E' -m 'Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>'
[hotfix 708ec94] fix: correct assignment number in README
1 file changed, 1 insertion(+), 1 deletion(-)

$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

$ git merge --no-ff hotfix -m "Merge branch 'hotfix'" -m
'Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>'
Merge made by the 'ort' strategy.
 README.md | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)

$ git branch -d hotfix
Deleted branch hotfix (was 708ec94).
```

```
|/
* 33fa36b Initial Commit
```

git rebase main on dev replayed dev's 11 commits on top of the hotfix merge. The README commit from A4 conflicted, because both sides changed line 1. I resolved it by keeping dev's expanded README with the hotfix's corrected number, then continued the rebase.

```
git rebase main (conflict)

$ git switch dev
Switched to branch 'dev'
Your branch is ahead of 'origin/dev' by 3 commits.
(use "git push" to publish your local commits)

$ git rebase main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply eb24041... docs: expand README with
pipeline overview and setup
hint: Resolve all conflicts manually, mark them as resolve
d with
hint: "git add/rm <conflicted_files>", then run "git rebas
e --continue".
hint: You can instead skip this commit: run "git rebase --
skip".
hint: To abort and get back to the state before "git rebas
e", run "git rebase --abort".
Could not apply eb24041... docs: expand README with pipeli
ne overview and setup

$ git status --short
UU README.md

$ head -8 README.md
<<<<<<< HEAD
## MLOps Assignment 3
=====
## MLOps Assignment 1: Fashion-MNIST ANN Pipeline

A fully connected neural network that classifies Fashion-M
NIST images into 10 clothing categories,
versioned end to end with Git and DVC.
```

```
Resolution

$ head -3 README.md
## MLOps Assignment 3: Fashion-MNIST ANN Pipeline

A fully connected neural network that classifies Fashion-M
NIST images into 10 clothing categories,

$ git add README.md

$ git rebase --continue
[detached HEAD 9f867b5] docs: expand README with pipeline
overview and setup
1 file changed, 23 insertions(+), 1 deletion(-)
Successfully rebased and updated refs/heads/dev.

$ git status --short
```

After rebase

```
$ git log --oneline --graph --all
* a7ab962 chore: add scratch script to inspect raw arrays
* 2c98562 feat: assert normalized pixels stay within [0, 1]
* 9f867b5 docs: expand README with pipeline overview and setup
* 5879d42 feat: add evaluate.py, metrics.json, confusion_matrix.png
* b986e3c feat: create train.py
* 6599485 chore: modified params.yaml opening structure
* 2677bd3 feat: add params.yaml and preprocess.py
* 2598fdb chore: changing keras declaration
* 5532086 feat: update .gitignore
* 51db866 feat: setup requirements.txt
* 96c2748 feat: create prepare.py
* bdd6d6f Merge branch 'hotfix'
| \
| * 708ec94 fix: correct assignment number in README
|/
| * 85e40dd feat: add evaluate.py, metrics.json, confusion_matrix.png
| * bcbf0cc feat: create train.py
| * 8e2f0da chore: modified params.yaml opening structure
```

```
| * 6e3f3a6 feat: add params.yaml and preprocess.py
| * 184b6fd chore: changing keras declaration
| * 9901c46 feat: update .gitignore
| * 7c201e3 feat: setup requirements.txt
| * 7d60d03 feat: create prepare.py
|/
* 33fa36b Initial Commit
```

After the rebase, `dev` sits in a straight line on top of `main`. Its commits got new hashes, so the old chain (`origin/dev`, right-hand side) remains until `dev` is force-pushed with `git push --force-with-lease origin dev`.

A7 Reset

On a `scratch` branch I made throwaway commits. `git reset --soft HEAD~1` removed commit 1 but left `throwaway1.txt` **staged** (A), ready to recommit. `git reset --hard HEAD~1` removed commit 2 **and** its file: the working tree is clean and `throwaway2.txt` no longer exists. Soft reset moves only the branch pointer. Hard reset also resets the staging area and working tree, discarding the changes (recoverable only through `git reflog`).

```
reset --soft

$ git switch -c scratch
Switched to a new branch 'scratch'

$ echo 'throwaway 1' > throwaway1.txt && git add throwaway
1.txt && git commit -q -m 'chore: throwaway commit 1' && g
it log --oneline -2
b5e4951 chore: throwaway commit 1
a7ab962 chore: add scratch script to inspect raw arrays

$ git reset --soft HEAD~1

$ git log --oneline -1
a7ab962 chore: add scratch script to inspect raw arrays

$ git status --short
A  throwaway1.txt

$ git diff --staged --stat
throwaway1.txt | 1 +
1 file changed, 1 insertion(+)
```

```
reset --hard

$ git commit -q -m 'chore: throwaway commit 1' && echo 'th
rowaway 2' > throwaway2.txt && git add throwaway2.txt && g
it commit -q -m 'chore: throwaway commit 2' && git log --o
neline -3
d26a06d chore: throwaway commit 2
b5e4951 chore: throwaway commit 1
a7ab962 chore: add scratch script to inspect raw arrays

$ git reset --hard HEAD~1
HEAD is now at b5e4951 chore: throwaway commit 1

$ git log --oneline -2
b5e4951 chore: throwaway commit 1
a7ab962 chore: add scratch script to inspect raw arrays

$ git status --short

$ ls throwaway2.txt
ls: cannot access 'throwaway2.txt': No such file or direct
ory
```

A8 git mv and git rm

The confusion-matrix image was moved from the project root into `reports/` with `git mv` (and `evaluate.py` updated to write there). A scratch inspection script that was no longer needed was removed with `git rm`. Git records the first as a rename (R, => in the stat), so the file's history is kept. It records the second as a deletion (D). Both were staged in one step, with no manual delete and re-add.

```
git mv

$ mkdir -p reports && git mv confusion_matrix.png reports/
confusion_matrix.png

$ git status --short
R  confusion_matrix.png -> reports/confusion_matrix.png

$ git status --short
M  README.md
R  confusion_matrix.png -> reports/confusion_matrix.png
M  src/evaluate.py

$ git commit -q -m 'refactor: move confusion matrix into r
eports/' -m 'Co-Authored-By: Claude Opus 5.5 <noreply@anth
ropic.com>' && git show --stat --format='%h %s' HEAD
04b31dd refactor: move confusion matrix into reports/

    README.md                                |
2 +-
confusion_matrix.png => reports/confusion_matrix.png | Bi
n
src/evaluate.py                             |
6 +++++
3 files changed, 5 insertions(+), 3 deletions(-)
```

```
git rm

$ git rm scratch_inspect.py
rm 'scratch_inspect.py'

$ git status --short
D  scratch_inspect.py

$ git commit -q -m 'chore: remove obsolete scratch script'
-m 'Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com
>' && git show --stat --format='%h %s' HEAD
ac35d60 chore: remove obsolete scratch script

    scratch_inspect.py | 6 -----
1 file changed, 6 deletions(-)

$ ls scratch_inspect.py
ls: cannot access 'scratch_inspect.py': No such file or di
rectory
```

Final state after Part A

Final history

```
$ git log --oneline --graph --all --decorate
* ac35d60 (HEAD -> dev) chore: remove obsolete scratch script
* 04b31dd refactor: move confusion matrix into reports/
| * b5e4951 (scratch) chore: throwaway commit 1
|/
* a7ab962 chore: add scratch script to inspect raw arrays
* 2c98562 feat: assert normalized pixels stay within [0, 1]
* 9f867b5 docs: expand README with pipeline overview and setup
* 5879d42 feat: add evaluate.py, metrics.json, confusion_matrix.png
* b986e3c feat: create train.py
* 6599485 chore: modified params.yaml opening structure
* 2677bd3 feat: add params.yaml and preprocess.py
* 2598fdb chore: changing keras declaration
* 5532086 feat: update .gitignore
* 51db866 feat: setup requirements.txt
* 96c2748 feat: create prepare.py
* bdd6d6f (main) Merge branch 'hotfix'
| \
| * 708ec94 fix: correct assignment number in README
|/
| * 85e40dd (origin/dev) feat: add evaluate.py, metrics.json, confusion_matrix.png
| * bcbf0cc feat: create train.py
| * 8e2f0da chore: modified params.yaml opening structure
| * 6e3f3a6 feat: add params.yaml and preprocess.py
| * 184b6fd chore: changing keras declaration
| * 9901c46 feat: update .gitignore
| * 7c201e3 feat: setup requirements.txt
| * 7d60d03 feat: create prepare.py
|/
* 33fa36b (origin/main, origin/HEAD) Initial Commit

$ git status --short
```